Enriching Scientific Knowledge Graph with Entropy-driven Progressive Self-Feedback Fusion

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***Abstract*—This technical report provides additional details on the method, dataset tasks and complete experimental results for the paper "Enriching Scientific Knowledge Graph with Entropy-driven Progressive Self-Feedback Fusion".**

***Index Terms*—scientific knowledge graph, scientific knowledge graph fusion, entropy-driven progressive self-feedback**

TABLE I
STATISTICS OF THE SciFusion-Bench. "$\#F_{fused}$": FUSED FACTS FROM GKG TO SKG. "*Str. Sim.*": AVERAGE ENTITY NEIGHBOR STRUCTURE SIMILARITY [1]. "*#Rel. O.*": THE PROPORTION OF OVERLAPPING RELATIONS IN SKG AND GKG.

Cluster	Dataset	Domain	$\#F_{fused}$	Str. Sim. ⇓	Rel. O. ⇓
LS	SKGF(W-Bio)	Biomedical	12,430	8.2%	0%
	SKGF(W-Plant)	Botany	8,120	13.2%	0%
NS	SKGF(W-Mat)	Materials	11,440	5.7%	0%
SHS	DKGF(W-I)	Politics	796,254	15.4%	0%
	DKGF(Y-I)	Politics	451,158	14.0%	0%
	SKGF(W-Music)	Musicology	5,057	16.8%	0%

I. EXPERIMENTS

To systematically evaluate the performance, architectural robustness, and processing efficiency of Self-Fusion, we conduct empirical investigations designed to address the following core research questions:

- **RQ1: Does Self-Fusion outperform established general-purpose and cross-task adaptation benchmark configurations across multi-domain scientific contexts?**
- **RQ2: What is the effectiveness of each component within Self-Fusion?**
- **RQ3: Does the Self-Fusion successfully balance accuracy, efficiency, and scalability?**
- **RQ4:How sensitive and robust is Self-Fusion under variations of primary parameters?**
- **RQ5: How compatible and adaptable is Self-Fusion across different foundation models and more complex scenarios?**
- **RQ6: Can Self-Fusion accurately identify domain-specific entities, and how does it operate in a concrete scenario?**
- **RQ7: Can the enhanced scientific knowledge graphs generated by Self-Fusion effectively boost performance in complex downstream scientific reasoning tasks?**

A. Experimental Settings

1) Benchmark Datasets: To evaluate the robustness of Self-Fusion across diverse scientific disciplines, we construct SciFusion-Bench, a multi-domain benchmark encompassing six standardized datasets. As illustrated in Table I, each dataset follows the prior criteria [2], where general knowledge from Wikipedia (via Wikidata or YAGO) is selectively integrated into a specialized SKG. The benchmark is categorized into three major scientific clusters:

- **Life Science (LS).** SKGF(W-Bio) integrates general molecular attributes and genetic metadata from Wikidata into PrimeKG [3], a precision medicine KG. SKGF(W-Plant) fuses Wikipedia's botanical descriptions into the specialized taxonomy and trait records of AgroLD [4].
- **Physical Science (PS).** SKGF(W-Mat) identifies chemical nomenclature and physical constants from Wikidata to enrich the experimental crystal structure data in the materials project [5].
- **Social & Humanistic Science (SHS).** It includes DKGF(W-I) and DKGF(Y-I), which selectively fuse background entity facts from Wikidata/YAGO into the political event streams of ICEWS (proposed by [2]). SKGF(W-Music) integrates artist biographies from Wikipedia into the musicology structures of MusicBrainz [6], [7].

Following the construction protocol of [2], each dataset bridges a domain-specific SKG with a general-purpose GKG (Wikidata or YAGO) by aligning entities across knowledge sources and mapping GKG predicates to domain-specific SKG relations. Candidate fusion facts are verified against curated source databases to obtain gold labels, with balanced positive and negative samples. The benchmarks exhibit high heterogeneity with low structural similarity and fully disjoint relation sets between SKG and GKG, forming a robust testbed for cross-domain fusion. All datasets adopt an S1 (80/10/10%) split as the primary setting; DKGF(W-I) and DKGF(Y-I) additionally provide an S2 (70/15/15%) configuration

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TABLE II

MAIN RESULTS ON DKGF(W-I) AND DKGF(Y-I). THE BEST ACC/F1 RESULTS ARE HIGHLIGHTED IN *bold*, WHILE THE RUNNER-UP RESULTS ARE *underlined*. " *Sema.*, *Struc.*, *LLM.*" INDICATE THE USE OF SEMANTICS, STRUCTURAL INFORMATION, AND LLMS, RESPECTIVELY.

Benchmark Configurations		Settings			DKGF(W-I)-S1				DKGF(Y-I)-S1				DKGF(W-I)-S2				DKGF(Y-I)-S2			
		Sema.	*Struc.*	*LLM.*	ACC	P	R	F1	ACC	P	R	F1	ACC	P	R	F1	ACC	P	R	F1
General-purpose																				
Rule.	StringMatch-F	✓			0.496	0.160	0.002	0.004	0.498	0.286	0.003	0.005	0.498	0.192	0.002	0.003	0.498	0.308	0.003	0.005
	TF-IDF-F	✓			0.498	0.461	0.025	0.047	0.507	0.598	0.043	0.081	0.498	0.451	0.023	0.043	0.508	0.598	0.046	0.086
Trans.	TransE-F	✓	✓		0.649	0.668	0.596	0.630	0.588	0.782	0.244	0.372	0.637	0.662	0.560	0.607	0.575	0.772	0.212	0.332
	TransH-F	✓	✓		0.640	0.683	0.520	0.591	0.583	0.764	0.240	0.365	0.632	0.685	0.489	0.570	0.568	0.768	0.195	0.311
	DistMult-F	✓	✓		0.554	0.547	0.629	0.585	0.525	0.524	0.548	0.536	0.536	0.532	0.590	0.560	0.532	0.530	0.562	0.545
	ComplEx-F	✓	✓		0.553	0.546	0.632	0.586	0.526	0.525	0.567	0.545	0.544	0.539	0.609	0.572	0.515	0.514	0.546	0.530
GNN.	GCN-F		✓		0.485	0.487	0.572	0.526	0.501	0.501	0.793	0.614	0.493	0.494	0.582	0.534	0.498	0.499	0.796	0.613
	TransGNN-F	✓	✓		0.490	0.491	0.505	0.498	0.465	0.465	0.477	0.471	0.486	0.487	0.507	0.497	0.468	0.468	0.481	0.474
	Graph-Memba-F	✓	✓		0.485	0.487	0.572	0.526	0.503	0.502	0.794	0.615	0.493	0.494	0.582	0.534	0.499	0.501	0.796	0.615
Generative.	BERT-F	✓			0.532	0.531	0.549	0.540	0.586	0.599	0.523	0.558	0.530	0.531	0.507	0.519	0.569	0.561	0.631	0.594
	ICL-F	✓		✓	0.550	0.553	0.528	0.540	0.583	0.573	0.649	0.609	0.494	0.494	0.523	0.508	0.488	0.489	0.523	0.505
	Self-Consistency-F	✓	✓	✓	0.592	0.607	0.522	0.561	0.590	0.584	0.621	0.602	0.531	0.533	0.507	0.520	0.566	0.559	0.628	0.591
	Self-RAG-F	✓	✓	✓	0.576	0.589	0.506	0.544	0.594	0.623	0.477	0.540	0.544	0.544	0.545	0.545	0.577	0.595	0.481	0.532
Cross-task Adaptation																				
EA.	SimpleHHEA-F	✓	✓		0.490	0.492	0.507	0.499	0.493	0.493	0.503	0.498	0.492	0.492	0.507	0.500	0.477	0.477	0.481	0.479
	ChatEA-F	✓	✓	✓	0.596	0.551	0.581	0.566	0.649	0.727	0.477	0.576	0.592	0.611	0.507	0.554	0.592	0.619	0.481	0.541
KGC.	KG-BERT-F	✓	✓		0.523	0.522	0.546	0.534	0.516	0.515	0.531	0.523	0.507	0.507	0.515	0.511	0.503	0.503	0.481	0.492
	KG-LLaMA-F	✓	✓	✓	0.512	0.512	0.531	0.521	0.539	0.537	0.558	0.547	0.529	0.527	0.529	0.528	0.521	0.521	0.514	0.517
	KoPA-F	✓	✓	✓	0.558	0.556	0.577	0.566	0.561	0.556	0.602	0.578	0.549	0.547	0.564	0.556	0.541	0.543	0.522	0.532
	PRGC-F	✓	✓		0.503	0.503	0.508	0.505	0.507	0.507	0.515	0.511	0.504	0.504	0.507	0.506	0.503	0.503	0.481	0.492
RTE.	NoGen-BART-F	✓	✓		0.516	0.515	0.531	0.523	0.511	0.511	0.523	0.517	0.508	0.508	0.517	0.512	0.509	0.509	0.491	0.500
	NoGen-T5-F	✓	✓		0.510	0.510	0.522	0.516	0.507	0.507	0.515	0.511	0.509	0.509	0.519	0.514	0.504	0.504	0.509	0.506
D.	ExeFuse	✓	✓		<u>0.680</u>	0.673	0.708	<u>0.690</u>	<u>0.661</u>	0.615	0.717	<u>0.662</u>	<u>0.655</u>	0.627	0.683	<u>0.654</u>	<u>0.633</u>	0.634	0.682	<u>0.657</u>
	Self-Fusion (Ours)	✓	✓	✓	**0.779**	0.877	0.649	**0.746**	**0.750**	0.815	0.647	**0.721**	**0.753**	0.878	0.586	**0.703**	**0.691**	0.664	0.772	**0.714**

following [2]. To ensure reproducibility, the whole datasets can be found in the public repository[1].

2) Benchmark Configurations: Since SKGF is a nascent task with no pre-existing specialized scientific solutions, a primary contribution of this work is to establish a comprehensive and standardized evaluation suite to facilitate future research. Following [2], we systematically adapted 22 representative methods from related fields (e.g., KGC, EA) to the SKGF setting using a unified fusion scoring protocol. These configurations are categorized as follows:

- **General-purpose Configurations.** To establish benchmark performance using universal architectures, this category employs broadly applicable methods (e.g., TransE, GNN, BERT, LLMs) rather than task-specific designs. This category of configurations is mainly based on current advanced and classic general methods, including rule-based methods (i.e., "Rule."), such as StringMatch-F [8] and TF-IDF-F [9]; Translation-based methods (i.e., "Trans.") such as TransE-F [10], TransH-F [11], DistMult-F [12], and ComplEx-F [13]; GNN-based methods (i.e., "GNN") like GCN-F [14], [15], TransGNN-F [16], and Graph-Memba-F [17]; Generative methods (i.e., "Generative.") like BERT-F [18], ICL-F [19], [20], Self-Consistency-F [21], [22], and Self-RAG-F [23];
- **Cross-task Adaptation Configurations.** This category of configuration mainly involves improving representative methods from current related research tasks to adapt to the DKGF task, including entity alignment (i.e., "EA."), such as SimpleHHEA-F [1] and ChatEA-F [24]; knowledge graph completion (i.e., "KGC."), such as KG-BERT-F [25], [26], KG-LLaMA-F [26], [27], KoPA-F [26] and PRGC-F [28]; and relation triple extraction (i.e., "RTE."), like NoGen-BART-F [29], and NoGen-T5-F [29]; and domain-specific knowledge graph fusion (i.e., "D."), like ExeFuse [2].

[1]https://github.com/eduzrh/SKGF

3) Evaluation Protocol and Implementation Details: We evaluate SKGF quality via triple classification [26], [29] on label-balanced test sets, reporting Accuracy (ACC), Precision (P), Recall (R), and F1-score (F1). Following [2], we adopt two data split settings: S1 (80% training) and S2 (70% training). Other datasets in `SciFusion-Bench` follow the S1 split by default. All LLMs reported in Table II and ablation study are implemented using the same model version, gpt-4o. For subsequent experiments, unless otherwise specified, gpt-4o-mini is used as the default LLM configuration due to its lower computational cost. Runtime (in seconds) is reported for efficiency analysis.

B. Extended Multi-Domain Results

We provides the complete evaluation across all six `SciFusion-Bench` datasets. Table III reports ACC and F1 for representative baselines and Self-Fusion under the S1 split.

Life Science (LS). Self-Fusion consistently outperforms all baselines on both `SKGF(W-Bio)` and `SKGF(W-Plant)`, achieving the largest absolute gains over the runner-up ExeFuse. The advantage can be attributed to two aspects. On the one hand, LLM-based generative methods attain competitive recall but suffer high precision loss on LS domains due to hallucinated biological pathways; Self-Fusion's entropy bounding explicitly suppresses such high-uncertainty fusions, ensuring that only scientifically verified facts enter the SKG. On the other hand, embedding-based methods fail to resolve

TABLE III
FULL EVALUATION ACROSS ALL SIX `SciFusion-Bench` DATASETS (S1 SPLIT). BEST RESULTS ARE **BOLDED**; RUNNER-UP RESULTS ARE UNDERLINED. `DKGF(W-I)`/`DKGF(Y-I)` RESULTS ARE CONSISTENT WITH TABLE II. "*Improv.*": RELATIVE IMPROVEMENT OF SELF-FUSION OVER THE RUNNER-UP (EXEFUSE) ON ACC AND F1 RESPECTIVELY.

Cat.	Method	`SKGF(W-Bio)` (Biomed.)		`SKGF(W-Plant)` (Botany)		`SKGF(W-Mat)` (Materials)		`SKGF(W-Music)` (Arts)		`DKGF(W-I)` (Politics)		`DKGF(Y-I)` (Politics)	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
	General-purpose Configurations												
Rule.	StringMatch-F	0.498	0.005	0.499	0.006	0.497	0.004	0.501	0.008	0.496	0.004	0.498	0.005
	TF-IDF-F	0.509	0.058	0.514	0.067	0.505	0.051	0.521	0.074	0.498	0.047	0.507	0.081
Trans.	TransE-F	0.618	0.597	0.633	0.614	0.591	0.572	0.641	0.622	0.649	0.630	0.588	0.372
	TransH-F	0.582	0.565	0.605	0.592	0.554	0.548	0.612	0.601	0.640	0.591	0.583	0.365
	DistMult-F	0.548	0.531	0.562	0.549	0.533	0.524	0.571	0.557	0.554	0.585	0.525	0.536
	ComplEx-F	0.545	0.529	0.559	0.547	0.530	0.521	0.568	0.554	0.553	0.586	0.526	0.545
GNN.	GCN-F	0.545	0.530	0.570	0.561	0.521	0.515	0.588	0.575	0.485	0.526	0.501	0.614
	TransGNN-F	0.519	0.503	0.538	0.526	0.505	0.496	0.551	0.541	0.490	0.498	0.465	0.471
	Graph-Memba-F	0.534	0.521	0.558	0.542	0.516	0.507	0.576	0.566	0.485	0.526	0.503	0.615
Gen.	BERT-F	0.541	0.524	0.558	0.545	0.524	0.514	0.573	0.561	0.532	0.540	0.586	0.558
	ICL-F	0.558	0.541	0.574	0.562	0.541	0.530	0.589	0.577	0.550	0.540	0.583	0.609
	Self-Consistency-F	0.601	0.584	0.619	0.603	0.573	0.558	0.632	0.617	0.592	0.561	0.590	0.602
	Self-RAG-F	0.625	0.608	0.645	0.635	0.602	0.598	0.665	0.658	0.576	0.544	0.594	0.540
	Cross-task Adaptation Configurations												
EA.	SimpleHHEA-F	0.503	0.488	0.518	0.508	0.496	0.487	0.528	0.518	0.490	0.499	0.493	0.498
	ChatEA-F	0.642	0.635	0.678	0.669	0.625	0.618	0.692	0.685	0.596	0.566	0.649	0.576
KGC.	KG-BERT-F	0.531	0.516	0.546	0.535	0.514	0.504	0.558	0.548	0.523	0.534	0.516	0.523
	KG-LLaMA-F	0.548	0.532	0.562	0.551	0.530	0.520	0.575	0.565	0.512	0.521	0.539	0.547
	KoPA-F	0.574	0.558	0.590	0.579	0.556	0.546	0.604	0.594	0.558	0.566	0.561	0.578
	PRGC-F	0.516	0.500	0.530	0.519	0.502	0.493	0.540	0.530	0.503	0.505	0.507	0.511
RTE.	NoGen-BART-F	0.524	0.508	0.539	0.528	0.507	0.498	0.548	0.538	0.516	0.523	0.511	0.517
	NoGen-T5-F	0.519	0.503	0.533	0.522	0.502	0.494	0.542	0.532	0.510	0.516	0.507	0.511
D.	ExeFuse	<u>0.685</u>	<u>0.692</u>	<u>0.715</u>	<u>0.724</u>	<u>0.688</u>	<u>0.695</u>	<u>0.718</u>	<u>0.725</u>	<u>0.680</u>	<u>0.690</u>	<u>0.661</u>	<u>0.662</u>
	Self-Fusion (Ours)	**0.768**	**0.775**	**0.795**	**0.802**	**0.774**	**0.786**	**0.755**	**0.762**	**0.779**	**0.746**	**0.750**	**0.721**
	Improv.	*+12.1%*	*+12.0%*	*+11.2%*	*+10.8%*	*+12.5%*	*+13.1%*	*+5.2%*	*+5.1%*	*+14.6%*	*+8.1%*	*+13.5%*	*+8.9%*

the fine-grained relational specificity required for biomedical and botanical knowledge, whereas Self-Fusion's structural fuzzy perception captures the topological cues needed to map coarse GKG predicates onto rigorous domain-specific relations.

Physical Science (PS). Self-Fusion achieves the strongest improvement margin across all domains on `SKGF(W-Mat)` (materials science). Physical science presents the starkest granularity gap: a GKG may relate two materials by a coarse compositional predicate while the SKG demands exact crystallographic or symmetry-group relations. Self-Fusion's entropy-driven self-feedback loop directly addresses this gap by iteratively verifying candidate facts against the local scientific context, filtering out coarse-grained associations that do not satisfy domain-specific precision requirements. Embedding-based baselines, lacking this verification mechanism, degrade significantly when faced with the semantic distance between general and specialized scientific predicates.

Social & Humanistic Science (SHS). Self-Fusion maintains consistent superiority on `SKGF(W-Music)` and the large-scale political datasets, though the improvement margin on `SKGF(W-Music)` is the smallest across all domains. This is consistent with the observation that social-science entities carry richer and more consistent surface descriptions in GKGs, partially reducing the entropy that our filter is designed to resolve. On the large-scale political datasets, where GKGs contain a high volume of ambiguous and noisy event-centric facts, Self-Fusion's entropy filter provides the largest benefit by selectively escalating only high-uncertainty candidates to LLM verification, suppressing noise that overwhelms baselines without such selective routing.

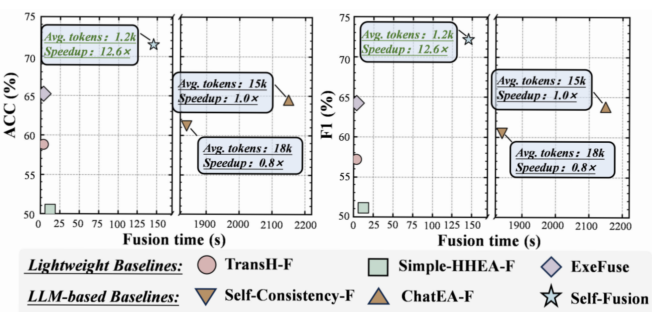

Fig. 1. Efficiency and scalability analysis on the AI4S dataset `SKGF(W-Mat)`. "*Speedup*": relative speed improvement compared to the LLM-based baseline.

TABLE IV
PERFORMANCE (F1-SCORE) W.R.T. TRAINING DATA RATIO ON `SKGF(W-Bio)`. "Δ" DENOTES IMPROVEMENT OVER THE BEST BASELINE.

Method	20%	40%	60%	80%	100%
TransH-F	0.352	0.415	0.488	0.542	0.565
GCN-F	0.320	0.395	0.462	0.510	0.530
ChatEA-F	0.485	0.542	0.589	0.620	0.635
Self-Fusion	**0.654**	**0.702**	**0.738**	**0.765**	**0.775**
Improvement (Δ)	*+34.8%*	*+29.5%*	*+25.3%*	*+23.4%*	*+22.0%*

C. Efficiency Analysis (RQ3)

Efficiency. Self-Fusion dominates LLM-based baselines in inference cost. As shown in Figure 1, Self-Fusion achieves

TABLE V

SCALABILITY UNDER PROPORTIONAL SKG+GKG SCALING ON SKGF (W-Bio) (1.2M–12M TOTAL FACTS): F1 AND TOTAL INFERENCE TIME (S). BEST F1 PER SCALE IS **BOLDED**; RUNNER-UP IS UNDERLINED.

Method	10% (1.2M)		20% (2.4M)		40% (4.8M)		60% (7.2M)		80% (9.6M)		100% (12M)	
	F1	Time	F1	Time	F1	Time	F1	Time	F1	Time	F1	Time
ICL-F	0.556	2,134	0.574	4,876	0.531	12,284	0.558	24,106	0.503	37,412	0.441	52,874
Self-Consistency-F	0.599	2,298	0.621	5,512	0.574	13,587	0.603	26,934	0.541	42,187	0.478	59,823
Self-RAG-F	0.623	2,261	0.647	5,238	0.598	12,842	0.629	25,312	0.563	39,548	0.498	55,487
ChatEA-F	<u>0.650</u>	2,688	<u>0.668</u>	5,064	<u>0.624</u>	9,887	<u>0.651</u>	14,823	<u>0.592</u>	18,674	<u>0.538</u>	26,928
Self-Fusion	**0.775**	**103.2**	**0.751**	**108.6**	**0.783**	**114.3**	**0.748**	**118.7**	**0.769**	**122.4**	**0.746**	**124.8**

TABLE VI

ACC AND AVERAGE TOKENS PER FACT UNDER PROPORTIONAL SCALING. BEST ACC PER SCALE IS **BOLDED**; RUNNER-UP IS UNDERLINED.

Method	10% (1.2M)		20% (2.4M)		40% (4.8M)		60% (7.2M)		80% (9.6M)		100% (12M)	
	ACC	Tokens	ACC	Tokens	ACC	Tokens	ACC	Tokens	ACC	Tokens	ACC	Tokens
ICL-F	0.561	12.0k	0.580	14.3k	0.537	17.8k	0.563	22.1k	0.508	25.6k	0.445	28.4k
Self-Consistency-F	0.610	15.0k	0.632	17.8k	0.581	21.4k	0.612	25.9k	0.548	29.3k	0.487	32.8k
Self-RAG-F	0.634	16.0k	0.657	18.9k	0.604	23.2k	0.638	27.8k	0.571	31.4k	0.507	34.5k
ChatEA-F	<u>0.657</u>	18.0k	<u>0.676</u>	19.6k	<u>0.631</u>	21.3k	<u>0.660</u>	22.8k	<u>0.599</u>	24.5k	<u>0.544</u>	26.2k
Self-Fusion	**0.768**	**1.2k**	**0.742**	**1.4k**	**0.776**	**1.6k**	**0.740**	**1.8k**	**0.762**	**2.0k**	**0.739**	**2.1k**

the optimal performance-to-cost ratio: it reduces token consumption by 93% and achieves a 12.6× speedup (145 s vs. 2,150 s) compared to ChatEA-F, while strictly outperforming it in accuracy (Avg. 0.719 vs. 0.642). The efficiency advantage can be attributed to two aspects. On the one hand, LLM-based methods invoke a full generative verification call per fact regardless of its complexity, which is computationally prohibitive at scale. On the other hand, Self-Fusion's entropy filter routes only genuinely high-uncertainty candidates to LLM verification, eliminating redundant inference on facts resolvable by lightweight structural matching. This selective activation effectively bridges the speed of embedding models with the reasoning depth of foundation models.

Performance in Low-Resource Settings. Scientific data is often scarce. We evaluated Self-Fusion and key baselines on SKGF(W-Bio) by varying the training data ratio from 20% to 100%. As shown in Table IV, Self-Fusion maintains high performance even with limited data. Notably, at 15.0% training data, Self-Fusion surpasses ChatEA-F (trained on 100% data), demonstrating that our *entropy-driven* mechanism learns generalized scientific logic rather than relying on memorization.

Large-Scale Scalability Analysis. To evaluate scalability under realistic graph size variation, we proportionally scale both the SKG and GKG of SKGF(W-Bio) from 10% to 100% of their full sizes, yielding a total fact volume ranging from 1.2M to 12M (SKG + GKG combined). We compare Self-Fusion against four LLM-based baselines at six scale points, reporting F1, total inference time (s), and average tokens per fact. All LLM-based methods use gpt-4o-mini. Times and token counts are anchored to the per-fact throughput measured on the base SKGF(W-Bio) and SKGF(W-Mat) experiments.

Tables V and VI depict the scalability of Self-Fusion compared to LLM-based baselines when varying the size of the background graph. Self-Fusion completes fusion within minutes across all scale points, while LLM-based baselines require hours at full scale. In addition, Self-Fusion's inference time grows far more slowly than the data volume, and it consistently maintains competitive F1 and ACC, whereas baselines degrade substantially in both efficiency and quality under scale pressure. The reasons can be concluded in two aspects. On the one hand, LLM-based methods invoke an independent full LLM call per fact, so as larger background graphs introduce longer context windows and more retrieval hops, both per-fact token consumption and total inference time grow super-linearly. On the other hand, Self-Fusion's entropy filter ensures that only a sub-linear fraction of facts is escalated to expensive LLM verification, keeping per-fact token consumption only marginally increased and total time growth well below the rate of data scaling. This efficiency in handling large-scale fusion further supports the framework's ability to operate effectively in real-world scientific knowledge graph scenarios.

II. RELATED WORK

Scientific Knowledge Graph. Scientific knowledge graphs have emerged as pivotal tools for organizing structured knowledge in specialized domains like biomedicine and chemistry [30], [31]. Unlike GKGs that prioritize broad coverage, SKGs are engineered for high-precision scientific reasoning and decision-making, such as protein-protein interaction prediction or crystal discovery [5], [32], [33]. However, the construction of SKGs is often hindered by the high cost of wetlab experiments and data scarcity, leading to inherent incompleteness [34]–[36]. To bridge this gap without compromising domain rigor, it is essential to integrate external knowledge from massive GKGs [2], [37], [38]. Our work addresses this by introducing a fusion paradigm that explicitly manages the "entropy gap" between general and scientific knowledge, ensuring the process adheres to the strict determinacy required by scientific domains.

Knowledge Integration. Existing integration methods predominantly fall into two paradigms: integration from external sources and internal structured reasoning, yet both fall short of the unique requirements for SKGF. Integration from external sources, such as relation triple extraction (RTE), entity linking (EL) [39], [40], and entity alignment (EA) [24], [41]–[44], focus on mining or aligning facts. However, distinct from SKGF, RTE operates on unstructured text liable to high-entropy noise, while EL is limited to text-to-node disambiguation. Although EA addresses cross-graph data, it is restricted to identifying equivalent entity pairs rather than discovering and fusing novel, domain-enhancing facts. On the other hand, internal reasoning methods like knowledge graph completion (KGC) [26], [30], [45] are confined to inferring missing links within a single closed-world graph. In contrast, SKGF aims to identify *missing* entities and generate *new* facts from external GKGs to actively expand coverage. Although DKGF [2] attempts cross-graph fusion, it typically treats the process as a static, one-shot matching task, failing to discern latent scientific cues from ambiguous GKG noise or ensure the deterministic knowledge required by scientific domains.

Entropy Theory for Data Management. Entropy measures system uncertainty [46] and is widely used for sample selection in active learning [47] or sharpening predictions in semi-supervised learning [48]. In KG research, entropy typically quantifies structural complexity or rule reliability [49], [50]. Distinguishing from these statistical views, we are the first to employ entropy as a quality metric for scientific knowledge. We model SKGF as a thermodynamic-like process of *entropy reduction*, where fuzzy general knowledge is progressively crystallized into precise scientific mechanisms through self-feedback.

III. Conclusion and Future Work

In this work, we explore SKGF, aiming to enrich specialized SKGs with massive general knowledge. We identify the scientific knowledge fusion rigor as the core challenge which requires isolating latent scientific signals from general-purpose noise and refining them into context-constrained, deterministic facts. To resolve this granularity mismatch, we proposed Self-Fusion, an entropy-theory-driven progressive self-feedback framework. By coordinating a fuzzy retriever for uncertainty preservation and a self-feedback verification module for systemic entropy reduction, Self-Fusion guarantees factual correctness and structural alignment during the fusion process. Furthermore, we established `SciFusion-Bench`, a multidisciplinary evaluation benchmark covering life, physical, and social sciences. In the future, we plan to extend our framework to support multi-modal scientific data.

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The authors employed GPT-5 (OpenAI) and Claude-4.5-sonnet (Anthropic) during the drafting phase to enhance the linguistic quality and grammatical consistency of the manuscript. These tools were used exclusively to optimize sentence structure, correct grammatical errors, and improve readability across all sections. All automated recommendations were thoroughly reviewed, modified, and validated by the human authors, who retain full accountability for the entire contents of this article.

References

[1] X. Jiang, C. Xu, Y. Shen, Y. Wang, F. Su, Z. Shi, F. Sun, Z. Li, J. Guo, and H. Shen, "Toward practical entity alignment method design: Insights from new highly heterogeneous knowledge graph datasets," in *Proceedings of the ACM on Web Conference 2024, WWW 2024, Singapore, May 13-17, 2024*, T. Chua, C. Ngo, R. Kumar, H. W. Lauw, and R. K. Lee, Eds. ACM, 2024, pp. 2325–2336. [Online]. Available: https://doi.org/10.1145/3589334.3645720

[2] R. Zhao, W. Zeng, W. Zhang, C. Chen, Z. Li, X. Zhao, and L. Chen, "Panning for gold: Expanding domain-specific knowledge graphs with general knowledge," 2026. [Online]. Available: https://arxiv.org/abs/2601.10485

[3] P. Chandak, K. Huang, and M. Zitnik, "Building a knowledge graph to enable precision medicine," *Scientific Data*, vol. 10, no. 1, p. 67, 2023.

[4] A. Venkatesan, G. Tagny Ngompe, N. E. Hassouni, I. Chentli, V. Guignon, C. Jonquet, M. Ruiz, and P. Larmande, "Agronomic linked data (agrold): A knowledge-based system to enable integrative biology in agronomy," *PLoS One*, vol. 13, no. 11, p. e0198270, 2018.

[5] A. Merchant, S. Batzner, S. S. Schoenholz, M. Aykol, G. Cheon, and E. D. Cubuk, "Scaling deep learning for materials discovery," *Nature*, vol. 624, no. 7990, pp. 80–85, 2023.

[6] P. Lisena, M. Achichi, P. Choffé, C. Cecconi, K. Todorov, B. Jacquemin, and R. Troncy, "Improving (re-) usability of musical datasets: An overview of the doremus project," *Bibliothek Forschung und Praxis*, vol. 42, no. 2, pp. 194–205, 2018.

[7] L. Yang, S. Zhou, J. Cheng, F. Zhang, J. Wan, S. Wang, and M. Lee, "DAEA: Enhancing entity alignment in real-world knowledge graphs through multi-source domain adaptation," in *Proceedings of the 31st International Conference on Computational Linguistics*, O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. D. Eugenio, and S. Schockaert, Eds. Abu Dhabi, UAE: Association for Computational Linguistics, Jan. 2025, pp. 5890–5901. [Online]. Available: https://aclanthology.org/2025.coling-main.393/

[8] Y. Du, Z. Chen, C. Jia, X. Gao, and Y. Jiang, "Out of length text recognition with sub-string matching," in *AAAI-25, Sponsored by the Association for the Advancement of Artificial Intelligence, February 25 - March 4, 2025, Philadelphia, PA, USA*, T. Walsh, J. Shah, and Z. Kolter, Eds. AAAI Press, 2025, pp. 2798–2806. [Online]. Available: https://doi.org/10.1609/aaai.v39i3.32285

[9] D. Paulsen, Y. Govind, and A. Doan, "Sparkly: A simple yet surprisingly strong TF/IDF blocker for entity matching," *Proc. VLDB Endow.*, vol. 16, no. 6, pp. 1507–1519, 2023. [Online]. Available: https://www.vldb.org/pvldb/vol16/p1507-paulsen.pdf

[10] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, and O. Yakhnenko, "Translating embeddings for modeling multi-relational data," *Adv Neural Inf Process Syst*, vol. 26, 2013.

[11] Z. Sun, Z. Deng, J. Nie, and J. Tang, "Rotate: Knowledge graph embedding by relational rotation in complex space," in *ICLR (Poster)*. OpenReview.net, 2019.

[12] B. Yang, W. Yih, X. He, J. Gao, and L. Deng, "Embedding entities and relations for learning and inference in knowledge bases," in *ICLR (Poster)*, 2015.

[13] T. Trouillon, J. Welbl, S. Riedel, É. Gaussier, and G. Bouchard, "Complex embeddings for simple link prediction," in *ICML*, ser. JMLR Workshop and Conference Proceedings, vol. 48. JMLR.org, 2016, pp. 2071–2080.

[14] Z. Chen, J. Liu, X. Chen, X. Wang, and W. Yin, "Rethinking the capacity of graph neural networks for branching strategy," in *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, Eds., 2024. [Online]. Available: http://papers.nips.cc/paper_files/paper/2024/hash/dff528ce3e1390c88f10bbf5e722a241-Abstract-Conference.html

[15] Z. Zhang, J. Wang, J. Ye, and F. Wu, "Rethinking graph convolutional networks in knowledge graph completion," in *WWW '22: The ACM Web Conference 2022, Virtual Event, Lyon, France, April 25 - 29, 2022*, F. Laforest, R. Troncy, E. Simperl, D. Agarwal, A. Gionis, I. Herman, and L. Médini, Eds. ACM, 2022, pp. 798–807. [Online]. Available: https://doi.org/10.1145/3485447.3511923

[16] P. Zhang, Y. Yan, X. Zhang, C. Li, S. Wang, F. Huang, and S. Kim, "Transgnn: Harnessing the collaborative power of transformers and graph neural networks for recommender systems," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, G. H. Yang, H. Wang, S. Han, C. Hauff, G. Zuccon, and Y. Zhang, Eds. ACM, 2024, pp. 1285–1295. [Online]. Available: https://doi.org/10.1145/3626772.3657721

[17] A. Behrouz and F. Hashemi, "Graph mamba: Towards learning on graphs with state space models," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2024, Barcelona, Spain, August 25-29, 2024*, R. Baeza-Yates and F. Bonchi, Eds. ACM, 2024, pp. 119–130. [Online]. Available: https://doi.org/10.1145/3637528.3672044

[18] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2019, Minneapolis, MN, USA, June 2-7, 2019, Volume 1 (Long and Short Papers)*, J. Burstein, C. Doran, and T. Solorio, Eds. Association for Computational Linguistics, 2019, pp. 4171–4186. [Online]. Available: https://doi.org/10.18653/v1/n19-1423

[19] V. Kothapalli, H. Firooz, and M. Sanjabi, "Cot-icl lab: A synthetic framework for studying chain-of-thought learning from in-context demonstrations," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, 2025, pp. 14 620–14 642. [Online]. Available: https://aclanthology.org/2025.acl-long.712/

[20] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., 2022. [Online]. Available: http://papers.nips.cc/paper_files/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html

[21] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," in *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. [Online]. Available: https://openreview.net/pdf?id=1PL1NIMMrw

[22] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 7, pp. 3580–3599, 2024. [Online]. Available: https://doi.org/10.1109/TKDE.2024.3352100

[23] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-rag: Learning to retrieve, generate, and critique through self-reflection," in *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. [Online]. Available: https://openreview.net/forum?id=hSyW5go0v8

[24] X. Jiang, Y. Shen, Z. Shi, C. Xu, W. Li, Z. Li, J. Guo, H. Shen, and Y. Wang, "Unlocking the power of large language models for entity alignment," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2024, Bangkok, Thailand, August 11-16, 2024*, L. Ku, A. Martins, and V. Srikumar, Eds. Association for Computational Linguistics, 2024, pp. 7566–7583. [Online]. Available: https://aclanthology.org/2024.acl-long.408

[25] L. Yao, C. Mao, and Y. Luo, "KG-BERT: BERT for knowledge graph completion," *CoRR*, vol. abs/1909.03193, 2019. [Online]. Available: http://arxiv.org/abs/1909.03193

[26] Y. Zhang, Z. Chen, L. Guo, Y. Xu, W. Zhang, and H. Chen, "Making large language models perform better in knowledge graph completion," in *Proceedings of the 32nd ACM International Conference on Multimedia, MM 2024, Melbourne, VIC, Australia, 28 October 2024 - 1 November 2024*, J. Cai, M. S. Kankanhalli, B. Prabhakaran, S. Boll, R. Subramanian, L. Zheng, V. K. Singh, P. César, L. Xie, and D. Xu, Eds. ACM, 2024, pp. 233–242. [Online]. Available: https://doi.org/10.1145/3664647.3681327

[27] L. Yao, J. Peng, C. Mao, and Y. Luo, "Exploring large language models for knowledge graph completion," in *2025 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2025, Hyderabad, India, April 6-11, 2025*. IEEE, 2025, pp. 1–5. [Online]. Available: https://doi.org/10.1109/ICASSP49660.2025.10889242

[28] X. Lv, Y. Lin, Y. Cao, L. Hou, J. Li, Z. Liu, P. Li, and J. Zhou, "Do pre-trained models benefit knowledge graph completion? A reliable evaluation and a reasonable approach," in *Findings of the Association for Computational Linguistics: ACL 2022, Dublin, Ireland, May 22-27, 2022*, S. Muresan, P. Nakov, and A. Villavicencio, Eds. Association for Computational Linguistics, 2022, pp. 3570–3581. [Online]. Available: https://doi.org/10.18653/v1/2022.findings-acl.282

[29] Y. Li, X. Zeng, Y. Zeng, and Y. Lin, "Enhanced packed marker with entity information for aspect sentiment triplet extraction," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, G. H. Yang, H. Wang, S. Han, C. Hauff, G. Zuccon, and Y. Zhang, Eds. ACM, 2024, pp. 619–629. [Online]. Available: https://doi.org/10.1145/3626772.3657734

[30] E. Y. Hu, S. Oleshko, S. Firmani, H. Cheng, Z. Zhu, M. Ulmer, M. Arnold, M. Colomé-Tatché, J. Tang, S. Xhonneux *et al.*, "Enhancing link prediction in biomedical knowledge graphs with biopathnet," *Nature Biomedical Engineering*, pp. 1–23, 2026.

[31] H. Wang, T. Fu, Y. Du, W. Gao, K. Huang, Z. Liu, P. Chandak, S. Liu, P. Van Katwyk, A. Deac *et al.*, "Scientific discovery in the age of artificial intelligence," *Nature*, vol. 620, no. 7972, pp. 47–60, 2023.

[32] S. Li, S. Wei, C. Huang, Y. Zhang, G. Zhang, and S. Sun, "Extracting and reconstructing knowledge in materials science literature using large language models," *Communications Materials*, 2026.

[33] N. Zhang, Z. Bi, X. Liang, S. Cheng, H. Hong, S. Deng, Q. Zhang, J. Lian, and H. Chen, "Ontoprotein: Protein pretraining with gene ontology embedding," in *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net, 2022. [Online]. Available: https://openreview.net/forum?id=yfe1VMYAXa4

[34] X. Lin, H. E, C. Xu, G. Zhou, H. Luo, T. Hu, F. Su, N. Li, and M. Sun, "TFLEX: temporal feature-logic embedding framework for complex reasoning over temporal knowledge graph," in *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., 2023. [Online]. Available: http://papers.nips.cc/paper_files/paper/2023/hash/e71a42c64851834013e2658b69d7fe93-Abstract-Conference.html

[35] B. Zhang and H. Soh, "Extract, define, canonicalize: An llm-based framework for knowledge graph construction," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, EMNLP 2024, Miami, FL, USA, November 12-16, 2024*, Y. Al-Onaizan, M. Bansal, and Y. Chen, Eds. Association for Computational Linguistics, 2024, pp. 9820–9836. [Online]. Available: https://doi.org/10.18653/v1/2024.emnlp-main.548

[36] B. Abu-Salih, "Domain-specific knowledge graphs: A survey," *J. Netw. Comput. Appl.*, vol. 185, p. 103076, 2021. [Online]. Available: https://doi.org/10.1016/j.jnca.2021.103076

[37] A. Khan, T. Wu, and X. Chen, "LLM+KG: data management opportunities in unifying large language models + knowledge graphs," in *Proceedings of Workshops at the 50th International Conference on Very Large Data Bases, VLDB 2024, Guangzhou, China, August 26-30, 2024*. VLDB.org, 2024. [Online]. Available: https://vldb.org/workshops/2024/proceedings/LLM+KG/LLM+KG-1.pdf

[38] D. Le, K. Zhao, M. Wang, and Y. Wu, "Graphlingo: Domain knowledge exploration by synchronizing knowledge graphs and large language models," in *40th IEEE International Conference on Data Engineering, ICDE 2024, Utrecht, The Netherlands, May 13-16, 2024*. IEEE, 2024, pp. 5477–5480. [Online]. Available: https://doi.org/10.1109/ICDE60146.2024.00432

[39] H. Chen, X. Shen, Q. Lv, J. Wang, X. Ni, and J. Ye, "SAC-KG: exploiting large language models as skilled automatic constructors for domain knowledge graph," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2024, Bangkok, Thailand, August 11-16, 2024*, L. Ku, A. Martins, and V. Srikumar, Eds. Association for Computational Linguistics, 2024, pp. 4345–4360. [Online]. Available: https://doi.org/10.18653/v1/2024.acl-long.238

[40] Y. Li, X. Zeng, Y. Zeng, and Y. Lin, "Enhanced packed marker with entity information for aspect sentiment triplet extraction," in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, G. H. Yang, H. Wang, S. Han, C. Hauff, G. Zuccon, and Y. Zhang, Eds. ACM, 2024, pp. 619–629. [Online]. Available: https://doi.org/10.1145/3626772.3657734

[41] Z. Zheng, H. Yuan, M. Chen, N. Jiang, H. Wang, and S. Wang, "Geokgm: A multimodal large language model for zero-shot knowledge graph completion in geospatial databases," *Proc. ACM Manag. Data*, vol. 3, no. 6, pp. 1–28, 2025. [Online]. Available: https://doi.org/10.1145/3769796

[42] Y. Wang, H. Sun, J. Wang, J. Wang, W. Tang, Q. Qi, S. Sun, and J. Liao, "Towards semantic consistency: Dirichlet energy driven robust multi-modal entity alignment," in *40th IEEE International Conference on Data Engineering, ICDE 2024, Utrecht, The Netherlands, May 13-16, 2024*. IEEE, 2024, pp. 3559–3572. [Online]. Available: https://doi.org/10.1109/ICDE60146.2024.00274

[43] X. Tian, Z. Sun, and W. Hu, "Generating explanations to understand and repair embedding-based entity alignment," in *40th IEEE International Conference on Data Engineering, ICDE 2024, Utrecht, The Netherlands, May 13-16, 2024*. IEEE, 2024, pp. 2205–2217. [Online]. Available: https://doi.org/10.1109/ICDE60146.2024.00175

[44] R. Zhao, W. Zeng, W. Zhang, X. Zhao, J. Tang, and L. Chen, "Towards temporal knowledge graph alignment in the wild," *CoRR*, vol. abs/2507.14475, 2025. [Online]. Available: https://doi.org/10.48550/arXiv.2507.14475

[45] S. Huang, P. Li, H. Wang, and Z. Chen, "Multi-chain fusion reasoning for knowledge graph link prediction," *Electronics*, vol. 14, no. 20, p. 4127, 2025.

[46] C. E. Shannon, "A mathematical theory of communication," *The Bell system technical journal*, vol. 27, no. 3, pp. 379–423, 1948.

[47] D. Yu, M. Li, W. Shi, and Q. Yu, "Evidential mixture machines: Deciphering multi-label correlations for active learning sensitivity," in *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, Eds., 2024. [Online]. Available: http://papers.nips.cc/paper_files/paper/2024/hash/cbb1fa8e7f515e796cda6621a703492f-Abstract-Conference.html

[48] J. He, J. Du, and W. Ma, "Preventing dimensional collapse in self-supervised learning via orthogonality regularization," in *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, Eds., 2024. [Online]. Available: http://papers.nips.cc/paper_files/paper/2024/hash/ad7922fd4650f8aba5d8b067e622ca84-Abstract-Conference.html

[49] Y. Wei, X. Yu, T. Pan, A. Li, and L. Du, "Structural entropy guided agent for detecting and repairing knowledge deficiencies in LLMs," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. [Online]. Available: https://openreview.net/forum?id=hTGqC1h8Ig

[50] J. Shi, Z. Cao, R. Ma, Y. Hu, J. Zhou, X. Li, L. Bai, L. He, and B. Zhang, "Dualresearch: Entropy-gated dual-graph retrieval for answer reconstruction," *arXiv preprint arXiv:2510.08959*, 2025.